Universidad Internacional del Ecuador (UIDE)

Computer Science School

Network Communication Technology

Title: *GTA1_Phase_1_Integrated Project*

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1. Scenario and scope

A technology company has opened a second office. Both sites must work as one network so that employees in either site can use the same internal chat system. Our team will plan, build and test the network that connects the two sites, and the chat application (Go client and server, with a FastAPI authentication service) will run on top of it. We follow the reference architecture of the project document.

1.1 The two sites

Site	Role	Users	Network equipment	Main services
Site A	Main office. Hosts the chat and authentication servers.	About 50 employees and up to 20 guests	Router R1 (Raspberry Pi with OpenWrt), switches SW-A1 and SW-A2 (Cisco)	Chat server, FastAPI, DHCP, NAT, firewall, OSPF
Site B	New office.	About 50 employees and up to 20 guests	Router R2 (Raspberry Pi with OpenWrt), switches SW-B1 and SW-B2 (Cisco)	DHCP, NAT, firewall, OSPF

1.2 Estimated devices

Both sites use the same four VLANs. The numbers include some room to grow and they define the size of each subnet in section 4.

VLAN	Name	Devices	Site A	Site B
10	Users	Employee computers, laptops and phones	50	50
20	Servers	Chat server, FastAPI server and future servers	20	10
30	Guests	Visitor devices, internet only	20	20
99	Administration	Routers, switches and administrator computers	10	10

1.3 Communication needs

Who talks	What they need	Why
Employees in both sites and the chat server	Send messages and read the history.	Core function of the chat (CU-02 and CU-03).
Employees in both sites and the authentication service	Log in and log out.	Access to the chat needs a valid token (CU-01 and CU-04).
Chat server and authentication service	Check that a token is valid.	Every message or history request is validated.
Router R1 and router R2	Exchange routes over the shared Wi-Fi link (192.168.100.0/24).	So each site knows how to reach the VLANs of the other site (OSPF).

Users and guests of each site	Reach the internet through their own site.	Each site has independent internet access with NAT.
Administrator	Manage routers and switches.	Only from the Administration VLAN.

1.4 Scope

This plan covers: two sites connected with OSPF between OpenWrt routers over a shared Wi-Fi link, VLANs with VLSM addressing from 10.10.0.0/16, 802.1Q trunks, Router-on-a-Stick, EtherChannel, RSTP, DHCP, NAT and firewall, and the tests that will prove they work.

It does not cover: high availability, load balancing, cloud services, SDN, IPv6 or routing protocols other than OSPF. Full configuration and implementation come in the next phases.

2. Network requirements

This table explains how we will address each technology of the reference architecture and which requirement of the project document it supports.

Technology	Requirement	How we will use it
OpenWrt	RF-NET-01	Each site has a Raspberry Pi running OpenWrt (R1 in Site A and R2 in Site B). It works as router, DHCP server, firewall and NAT, and runs FRRouting for OSPF. It also keeps monitoring and logs, which we will use as a source of usage data. Its port eth0 faces the local switches (trunk) and its wlan0 connects to the shared Wi-Fi network (WAN).
VLANs	RF-NET-02	Both sites use the same four VLANs: 10 Users, 20 Servers, 30 Guests and 99 Administration. Using the same IDs keeps both configurations equal, while each site has its own subnets.
VLSM	RF-NET-01, RF-NET-02	We split 10.10.0.0/16 into eight subnets, one per VLAN in each site, with a mask that fits the estimated devices (/26, /27 and /28). The table is in section 4. The Wi-Fi link between routers (192.168.100.0/24) comes from the reference architecture and is outside this block.
IEEE 802.1Q trunks	RF-NET-02	The link from eth0 of the router to SW-x1, and the links between SW-x1 and SW-x2, carry VLANs 10, 20, 30 and 99 with 802.1Q tags. The end devices (users, servers, guests and administration) connect to access ports of SW-x2, and each access port carries only one VLAN.
Router-on-a-Stick	RF-NET-02	eth0 of each router is divided into sub-interfaces eth0.10, eth0.20, eth0.30 and eth0.99. Each one is the gateway of its VLAN and routes the traffic between VLANs according to the firewall rules.
EtherChannel	RF-NET-04	In each site, SW-x1 and SW-x2 are joined by an EtherChannel of 4 x 1 Gbps with LACP (802.3ad). The reference architecture also shows an optional backup bundle with PAgP, which we keep as an extra if time allows.
RSTP	RF-NET-04	The Cisco switches run RSTP, so the redundant links between them do not create loops and the network recovers quickly if a link fails.

OSPF	RF-NET-03	FRRouting on R1 and R2 runs OSPFv2 in area 0 over wlan0 (R1 192.168.100.10 and R2 192.168.100.11). Each router announces its local VLAN subnets and does not announce a default route. The LAN sub-interfaces are passive.
DHCP	RF-NET-01	OpenWrt gives addresses per VLAN to Users and Guests, with its own pool in each site. Servers and administration devices use fixed IP addresses.
NAT	RF-SEC-03	Each router applies NAT (masquerade) on wlan0 so its VLANs reach the internet independently of the other site. Traffic between the 10.10.0.0/16 networks of both sites is not translated.
Firewall	RF-SEC-03	The OpenWrt firewall (nftables or iptables) has one zone per VLAN. Traffic between zones is blocked by default and only the rules of section 5.4 are allowed.

3. Preliminary topology

This section shows the physical and the logical topology of the network. Both diagrams follow the reference architecture of the project document.

3.1 Physical topology

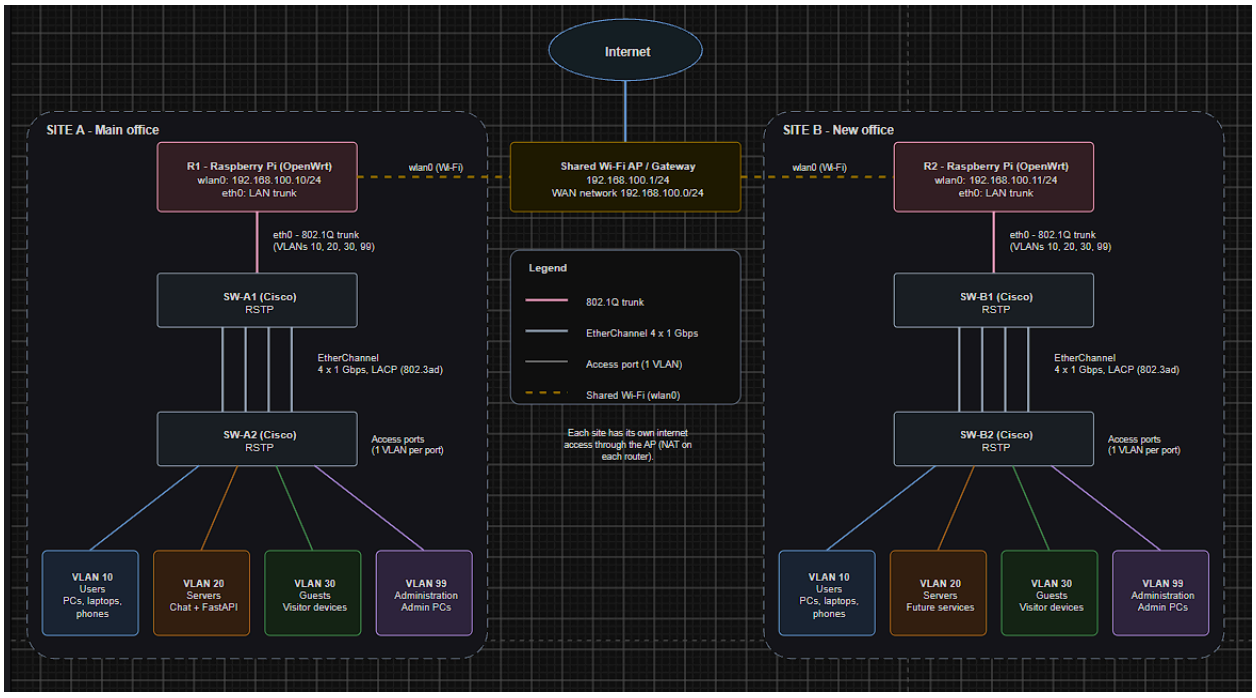
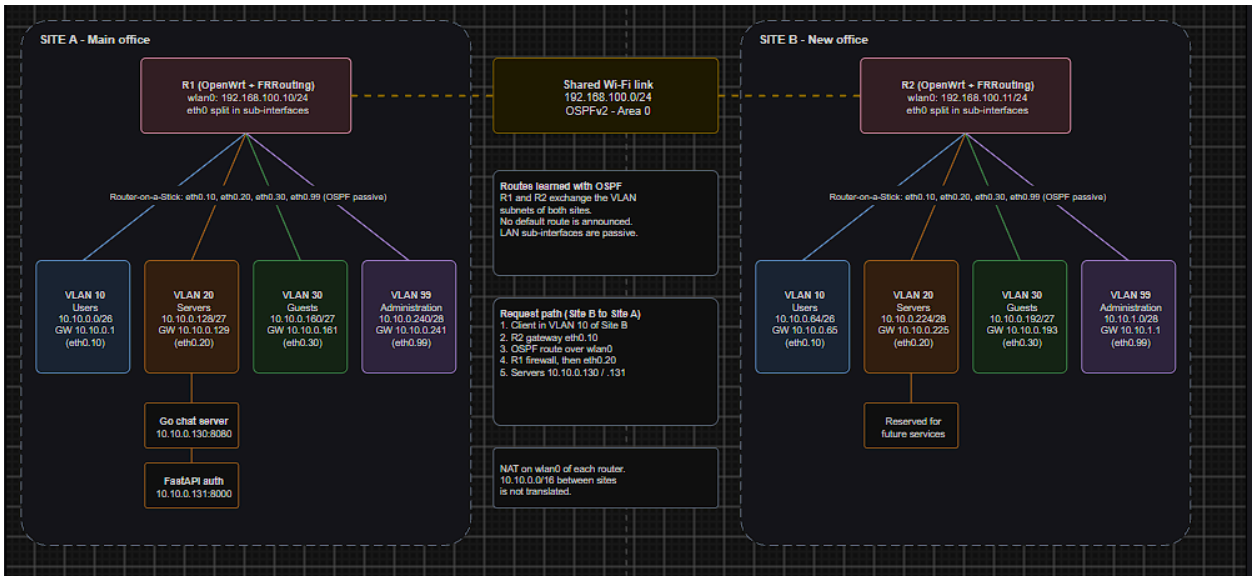


Figure 1 illustrates the links of the two locations. At both locations, Raspberry Pi router (R1 in Location A and R2 in Location B) links its wlan0 interface to the shared Wi-Fi access point and eth0 port to SW-x1 using the 802.1Q trunking method. SW-x1 and SW-x2 are connected through an EtherChannel of four 1 Gbps links using LACP and all the end devices (users, servers, guests and management) connect to access ports of SW-x2.

3.2 Logical topology



The logical design is presented in Figure 2. The two sites in question apply VLANs 10, 20, 30 and 99, with every router configured as the gateway for its VLANs via the sub-interfaces eth0.10, eth0.20, eth0.30 and eth0.99. The routers R1 (192.168.100.10) and R2 (192.168.100.11) share the VLAN subnets via the OSPFv2 in area 0 through a common Wi-Fi connection. The chat server and the FastAPI application are operating in VLAN 20, located at Site A, wherefore the users at Site B are able to access both services through the OSPF connection.

4. Addressing strategy

We will perform VLSM on the block 10.10.0.0/16. To do this, we first estimated the number of devices of each VLAN at each site (section 1.2). We sorted the eight subnets from the largest to the smallest and gave each one the smallest mask that can fit its devices so as not to waste addresses. The two Users VLANs, having 50 devices each, are allocated a /26 (62 usable hosts). The three VLANs with 20 devices (Servers of Site A and Guests of both sites) are allocated a /27 (30 usable hosts). The three VLANs with 10 devices (Servers of Site B and Administration of both sites) are allocated a /28 (14 usable hosts).

The various subnets do not overlap. They are numbered one after the other, starting from 10.10.0.0. The first usable address in each of the subnets will become the gateway, which will be a sub-interface of the router (eth0.10, eth0.20, eth0.30, or eth0.99). The chat server (10.10.0.130) and the FastAPI service (10.10.0.131) are given fixed addresses in the Servers VLAN of Site A (10.10.0.128/27), as outlined in section 5.

The eight subnets utilize only the initial 272 addresses of this block (10.10.1.15 is the last of these addresses). Hence, the unused portion of 10.10.0.0/16 that starts from 10.10.1.16 is reserved for future use. The project provides for the Wi-Fi network with the address of 192.168.100.0/24 outside this block.

Site	VLAN	Name	Devices	Network	Usable range	Gateway

A	10	Users	50	10.10.0.0/26	10.10.0.1 - 10.10.0.62	10.10.0.1
B	10	Users	50	10.10.0.64/26	10.10.0.65 - 10.10.0.126	10.10.0.65
A	20	Servers	20	10.10.0.128/27	10.10.0.129 - 10.10.0.158	10.10.0.129
A	30	Guests	20	10.10.0.160/27	10.10.0.161 - 10.10.0.190	10.10.0.161
B	30	Guests	20	10.10.0.192/27	10.10.0.193 - 10.10.0.222	10.10.0.193
B	20	Servers	10	10.10.0.224/28	10.10.0.225 - 10.10.0.238	10.10.0.225
A	99	Administration	10	10.10.0.240/28	10.10.0.241 - 10.10.0.254	10.10.0.241
B	99	Administration	10	10.10.1.0/28	10.10.1.1 - 10.10.1.14	10.10.1.1

5. Integration plan

The Go chat application and the FastAPI authentication service need the network to be ready before they can work. This section explains where each part will run and what traffic the network must allow.

5.1 Where each component will run

Component	Where	VLAN and IP	Port	Notes
Go client	User computers in Site A and Site B	VLAN 10, address from DHCP	None	Used by employees.
Go chat server	Site A, server host	VLAN 20, 10.10.0.130	TCP 8080	Stores messages and history.
FastAPI authentication	Site A, server host	VLAN 20, 10.10.0.131	TCP 8000	Creates and checks tokens.

We put both servers in the Servers VLAN of Site A, so every user in Site B reaches them through the OSPF link between the routers. This way the demonstration of RF-INT-01 also proves the connection between sites. The Servers VLAN of Site B is kept for future services. Each service keeps its own data on its own host. The IP addresses follow the addressing plan of section 4.

5.2 Path of a request from Site B

1. The client in VLAN 10 of Site B sends the request to 10.10.0.130 or 10.10.0.131.
2. Router R2 receives it on eth0.10, which is the gateway of that VLAN.

3. R2 finds the route to the Servers subnet of Site A, learned with OSPF, and sends it through wlan0 over the shared Wi-Fi network.
4. Router R1 receives it on wlan0, applies its firewall rules and sends it out through eth0.20 and the switches to the server.
5. The response returns along the same path.

The shared Wi-Fi link is the weakest point of the design, because it adds delay and can fail. We will measure it with the response time tests of RNF-01.

5.3 Communication between components

From	To	Operations	Protocol
Go client	FastAPI	POST /auth/login, POST /auth/logout	HTTP with JSON, TCP 8000
Go client	Go chat server	POST /api/mensajes, GET /api/mensajes/nuevos (every few seconds), GET /api/historial	HTTP with JSON, TCP 8080
Go chat server	FastAPI	GET /auth/validar	HTTP with JSON, TCP 8000

Every request to the chat server includes the token. If the link between the sites fails, the clients of Site B cannot reach the servers and show that they could not connect (alternative flow A2 of CU-01).

5.4 What the firewall must allow

#	From	To	Port	Action
1	Users (VLAN 10) of both sites	Servers (VLAN 20) of Site A	TCP 8080 and 8000	Allow
2	Administration (VLAN 99)	All VLANs	SSH 22 and HTTPS 443	Allow
3	R1 and R2 on wlan0	R1 and R2 on wlan0	OSPF (protocol 89)	Allow
4	Users and Guests	Internet	Any	Allow, with NAT
5	Guests (VLAN 30)	Users, Servers and Administration	Any	Deny
6	Users	Administration (VLAN 99)	Any	Deny
7	Any other traffic between zones	Any	Any	Deny

Because the sites are joined through wlan0, the traffic from the other site reaches each router from the WAN zone. Rule 1 must therefore allow the Users subnet of Site B arriving on wlan0 of R1, and the NAT must exclude destinations inside 10.10.0.0/16.

5.5 What has to be ready first

1. Switches with VLANs, trunks, EtherChannel and RSTP.
2. Routers with Router-on-a-Stick, so each VLAN has a gateway.
3. The Wi-Fi link and OSPF between R1 and R2, so the sites can reach each other.
4. DHCP, NAT and firewall rules.
5. The servers installed in VLAN 20 of Site A with the fixed IP addresses of 5.1.
6. The Go client in both sites, checking login, messages and history through the network.

6. Work plan

6.1 Activities, schedule, responsibilities and dependencies

Activity	Weeks	Owner	Depends on
Design	6-7	Cayetano (Network), Darwin (Go app & repo), José (Security & Stats)	None
Prototype	8-10	Cayetano, Darwin, José	Design
Integration	11-12	Cayetano, Darwin, José	Network (must be ready before integration)
Testing	13-14	Cayetano, Darwin, José	Integration (must be ready before tests)
Defense	15	Cayetano, Darwin, José	Testing
Closure	16	Cayetano, Darwin, José	Defense

6.2 Resources

Type	Required items
Hardware	2 Raspberry Pi, 4 Cisco switches, cables for EtherChannel, Wi-Fi AP, PCs, and 2 devices for the servers.
Software	OpenWrt, FRRouting, nftables, Go, Python with FastAPI, Git, ping, traceroute, Wireshark.
Implementation environment	Laboratory, or emulation environment if hardware is missing.

6.3 Technical risks

Risk	Proposed solution
Shared Wi-Fi is slow or unstable	Measure response times; use robust QoS policies or adjust OSPF timers if necessary.
OSPF is slow to form neighbors over Wi-Fi	Tune OSPF hello/dead intervals and ensure multicast traffic is passing correctly.
Firewall blocks OSPF by mistake	Explicitly allow OSPF (protocol 89) on wlan0 interfaces in the nftables ruleset.
LACP and PAgP cannot be mixed	Standardize on LACP (802.3ad) across all EtherChannel configurations.
Lack of hardware or time	Use emulation platforms and adhere strictly to the schedule.

7. Validation plan

Area	Test	Tool or command	Expected result	Evidence
Connectivity	Ping inside the same VLAN	ping	Successful ICMP echo replies	Screenshots / terminal logs
Connectivity	Ping between different VLANs and between sites	ping, traceroute	Successful routing between subnets and sites	Screenshots / terminal logs
Routing	Verify OSPF neighbors and routes	show ip ospf neighbor (FRRouting)	Routers R1 and R2 are full neighbors over wlan0	Screenshots / terminal logs
Redundancy	Check EtherChannel status after disconnecting 1 of 4 cables	show etherchannel summary	Bundle stays up with remaining links	Screenshots
Redundancy	Verify RSTP status	show spanning-tree	Correct root bridge and port roles	Screenshots

			(forwarding/blocking)	
Service access	Client from Site B logs in and chats with Server in Site A	Go client app	Successful login and message delivery	App screenshots / Wireshark captures
Security	Guest (VLAN 30) tries to reach servers (VLAN 20)	ping	Request times out (blocked by firewall)	Screenshots
Security	User (VLAN 10) tries to reach administration (VLAN 99)	ping, ssh	Request is denied (blocked by firewall)	Screenshots
Security	API request sent without a valid token	curl / Postman	HTTP 401 Unauthorized response	Screenshots / terminal logs
Security	Check database password storage	Database query	Only hashed passwords are stored, no plaintext	Screenshots
Security	Verify firewall ruleset	nft list ruleset	Expected firewall zones and rules are applied	Screenshots / logs

GitHub Repository Link:

https://github.com/cayetano-c/chat-empresarial